

PALERMO PUNTA RAISI, Italy

WMO: 164050

Lat: 38.182N

Lon: 13.099E

Elev: 112

StdP: 14.64

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	44.2	46.0	28.7	22.7	50.9	32.2	26.8	51.6	34.3	55.9	31.7	56.0	9.8	170	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.1	91.6	72.1	88.0	73.5	85.8	74.4	79.4	84.1	78.2	83.4	76.9	82.6	12.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.4	148.1	82.8	76.6	139.5	81.9	75.1	132.3	81.3	43.1	84.0	41.6	83.7	40.4	82.7	91.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.5	26.0	23.0	39.7	100.4	3.0	3.8	37.5	103.1	35.7	105.4	34.0	107.5	31.8	110.3		
(5)			36.1	81.6	2.5	3.0	34.3	83.7	32.9	85.5	31.5	87.1	29.7	89.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	65.8	54.4	53.9	56.6	60.7	66.6	73.7	78.8	79.9	75.1	69.4	62.9	57.2	(6)
(7)		DBStd	10.00	4.10	4.33	4.38	4.28	4.51	4.57	3.15	3.31	3.95	4.03	4.70	4.23	(7)
(8)		HDD50	23	8	10	3	0	0	0	0	0	0	0	0	2	(8)
(9)		HDD65	1425	330	311	265	144	29	0	0	0	0	7	92	247	(9)
(10)		CDD50	5806	145	120	208	321	516	712	891	927	753	601	388	224	(10)
(11)		CDD65	1732	1	1	5	15	80	262	426	462	303	143	30	4	(11)
(12)		CDH74	12763	0	5	37	71	300	1474	3808	4601	1925	450	89	3	(12)
(13)		CDH80	3234	0	0	9	9	89	362	955	1338	367	96	9	0	(13)
(14)	Wind	WSAvg	9.9	11.6	11.9	11.1	9.8	8.8	8.6	8.4	8.1	9.1	8.9	11.1	11.8	(14)
(15)	Precipitation	PrecAvg	24.5	4.3	3.1	2.9	2.6	1.1	0.5	0.2	0.7	3.2	4.3	4.4	3.9	(15)
(16)		PrecMax	39.9	18.0	15.2	15.8	6.4	4.4	1.5	0.7	3.5	14.8	17.0	13.0	6.6	(16)
(17)		PrecMin	15.6	0.6	0.3	0.0	0.3	0.1	0.0	0.0	0.0	0.3	0.0	1.1	2.6	(17)
(18)		PrecStd	5.4	4.3	3.2	3.4	1.5	1.1	0.4	0.2	0.9	3.5	4.4	3.1	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.1	73.2	78.8	80.9	91.4	96.6	95.0	97.2	92.8	89.2	81.0	73.6	(19)
(20)			MCWB	54.7	56.3	58.5	59.8	64.4	66.4	72.2	72.9	70.6	65.6	62.5	57.7	(20)
(21)		2%	DB	64.9	66.1	71.3	75.2	80.6	87.7	89.6	91.3	87.2	82.2	76.5	68.3	(21)
(22)			MCWB	55.3	54.9	57.3	59.8	64.9	69.9	72.5	74.4	71.8	67.9	61.7	57.7	(22)
(23)		5%	DB	62.6	62.6	65.9	70.1	76.6	83.7	86.3	87.8	83.8	78.4	73.1	65.6	(23)
(24)			MCWB	55.3	54.8	56.0	58.8	64.8	70.7	74.5	75.8	72.9	68.3	61.7	57.3	(24)
(25)		10%	DB	60.6	60.5	62.8	67.2	73.5	80.7	84.4	85.7	81.1	75.6	70.0	63.0	(25)
(26)			MCWB	54.7	54.3	56.0	58.8	64.9	71.2	74.7	76.1	72.1	67.9	61.6	56.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.4	59.2	61.2	64.3	70.2	77.9	80.6	82.1	78.6	74.4	67.5	61.5	(27)
(28)			MCD B	63.6	65.1	68.7	71.7	75.2	82.2	84.3	85.6	83.3	77.9	73.1	66.2	(28)
(29)		2%	WB	57.4	57.4	59.3	62.5	68.8	75.6	78.8	80.2	76.4	71.6	65.3	59.9	(29)
(30)			MCD B	61.9	62.3	64.7	67.8	73.8	80.6	83.8	84.7	81.7	76.7	71.2	65.0	(30)
(31)		5%	WB	56.2	56.0	57.8	61.1	67.4	74.1	77.5	78.9	74.8	69.9	63.7	58.6	(31)
(32)			MCD B	60.9	60.7	62.9	66.2	72.5	79.4	82.9	83.9	80.8	75.5	69.6	63.8	(32)
(33)		10%	WB	54.9	54.7	56.7	60.0	66.0	72.6	76.4	77.7	73.3	68.6	62.4	57.3	(33)
(34)			MCD B	59.9	59.4	61.6	65.1	71.3	78.1	82.1	83.3	79.8	74.4	68.2	62.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.0	9.6	10.6	11.1	11.2	11.4	11.2	11.1	10.3	10.7	9.9	8.8	(35)
(36)			MCDBR	10.5	12.2	14.9	16.6	16.7	16.8	14.9	14.9	13.4	13.8	12.9	10.7	(36)
(37)			MCWBR	7.5	8.0	8.5	9.0	8.7	9.5	10.1	10.5	9.5	8.3	8.3	7.1	(37)
(38)		5% WB	MCDBR	9.0	10.7	12.0	12.9	13.1	12.5	12.2	12.6	11.5	11.2	10.6	8.9	(38)
(39)			MCWBR	7.7	8.0	8.5	8.6	8.5	9.0	9.5	10.1	9.1	8.0	7.9	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.346	0.368	0.407	0.445	0.451	0.428	0.430	0.430	0.436	0.410	0.378	0.353	(40)	
(41)		taud	2.451	2.370	2.254	2.163	2.191	2.303	2.307	2.310	2.295	2.344	2.382	2.428	(41)	
(42)		Ebn at Noon	262	271	269	265	265	271	269	267	257	252	246	250	(42)	
(43)		Edh at Noon	27	33	40	47	46	41	41	40	38	33	28	26	(43)	
(44)	All-Sky Solar Radiation	RadAvg	692	956	1378	1782	2141	2392	2423	2123	1587	1156	792	619	(44)	
(45)		RadStd	50	93	113	104	109	72	53	96	71	76	49	52	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-1.51	-1.94	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	-1.01	-1.17	N/A	N/A	N/A	N/A

Nomenclature: See separate page